Tushar Tyagi

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Backend Developer with hands-on experience in **FastAPI, Django, Node.js, Express, Redis, Kafka, React.js** and scalable systems engineering. Skilled in **microservices, CI/CD, automation, caching layers, LLM workflows** and API-driven platforms.

Education

Guru Gobind Singh Indraprastha University — Dwarka, New Delhi
Bachelor of Technology (B.Tech) — *Aug 2019 – Aug 2023*
- Cumulative GPA: **8.4/10.0**

Work Experience

Backend Developer — Jul 2024 – Oct 2025
India Today Group — *Noida, India*
- Designed high-performance microservices using **FastAPI, Node.js, Express, Redis**.
- Built high-throughput APIs for **real-time analytics** with WebSockets and async workflows.
- Improved system latency by **60%** through Redis caching, optimized DB queries, and pooling.
- Developed CI/CD workflows using **Docker, GitHub Actions, AWS deployments**.
- Implemented monitoring and distributed logging using **AWS CloudWatch, NGINX load balancing**.

Software Developer — May 2023 – Jun 2024
Codalien Technologies — *New Delhi, India*
- Engineered backend services for analytics using **Express.js, Django, MongoDB**.
- Integrated **Kafka** for streaming telemetry pipelines and async event handling.
- Implemented **JWT/OAuth2 authentication**, rate-limited API gateways.
- Collaborated with frontend teams to integrate REST APIs with **React.js dashboards**.

Software Intern — Oct 2022 – Apr 2023
DS Group — *Noida, India*
- Developed Flask and Django modules for automation workflows and internal SaaS tools.
- Built reporting tools using **Pandas** and optimized SQL queries for large datasets.
- Created **Streamlit dashboards** with role-based access control.

Projects

Deep Research Pro — Python · FastAPI · Django · LangGraph · ChromaDB · Streamlit

- Developed an automated **LLM-powered research engine** generating structured analyst-grade reports.
- Implemented multi-agent workflow using **LangGraph + OpenRouter APIs**.
- Designed vector-based retrieval using **ChromaDB** with semantic routing and memory.
- Built interactive UI in **Streamlit** with PDF/Markdown export.

Live Stream Deduplication — Python · OpenCV · SSIM · Redis · Docker

- Developed real-time deduplication pipeline using **SSIM + OpenCV**.
- Boosted compute efficiency by **45%** using Redis caching and Bloom filters.
- Deployed using **NGINX reverse proxy, Docker Compose**, and monitoring agents.

YouTube Analytics Platform — Node.js · Express · MongoDB · React.js · AWS

- Developed backend APIs for content and audience analytics with scheduled aggregations.
- Implemented ingestion workers using **CRON + MongoDB pipelines**.
- Built real-time insight dashboards using React + Streamlit.

Technical Skills

Languages: Python, JavaScript, SQL, Bash
Backend: FastAPI, Django, Flask, Express.js, Node.js
Frontend: React.js, Streamlit
Databases: MongoDB, PostgreSQL, MySQL, Redis, ChromaDB
Messaging: Kafka, RabbitMQ, Celery
DevOps: Docker, AWS (EC2, Lambda, CloudWatch), GitHub Actions, Nginx
AI/ML: LLM Integration, RAG Pipelines, OpenCV, LangGraph, SSIM
Concepts: Microservices, Caching, Load Balancing, Authentication (JWT/OAuth2), System Design